TIPA Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information


| Proposer | Name | Joaquín Rodríguez-López (Lead)[1,2]<br>Nicholas E. Jackson[1,2] | Email | joaquinr@illinois.edu<br>jacksonn@illinois.edu |
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| Institution / Department | Beckman Institute for Advanced Science and Technology,[1] Department of Chemistry[2] | | | |
| Address | 405 N. Mathews Ave., Urbana, IL 61801;[1] 600 S. Mathews Ave, Urbana IL 61801.[2] | | | |


## 2. Technology Overview

| Title of Proposed Technology | ***Automated Multiplexed Electrochemistry for a Minimal Set Approach to the Electrocatalytic Conversion of Targeted Small Molecules*** | | | |
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| Korea's 12 National Strategic Technology Fields | Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (x), Cybersecurity ( ), AI (x), Next-gen Communication ( ), Advanced Robotics & Manufacturing (x), Quantum ( ) | | | |
| Industrial Technology Classification [1)] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | *Chemicals* | | *Mechanical-Materials* | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | *Chemical Processes* | *Catalyst Application Technologies* | *Robotics/Automation Machinery* | *Other Robotics/Automation Machinery Technologies* |
| Concept of Proposed Technology | We will use powerful automated electrochemical methods based on our Electrolab platform and informed by artificial intelligence methods for the electrodeposition of metal electrocatalysts resulting in the discovery, characterization, and prediction of optimized deposits for tackling targeted small molecule conversion based on a minimal set of descriptor reactions. By creating libraries of deposits at high throughput via programmable robotic electrochemistry, we will achieve a comprehensive understanding of their basic reactivity to core reactions, including hydrogen oxidation (HOR), hydrogen evolution (HER), oxygen evolution (OER), carbon monoxide oxidation (COR), methanol oxidation (MOR), and surface oxide/hydride quantification (SOHQ). This understanding will be used to 1) discover robust, statistically validated surfaces for these reactions using closed-loop Bayesian optimization (BO), and 2) use key descriptors of these reactions for artificial intelligence – | | | |

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| | led prediction of targeted small molecule electrocatalysis of interest to our industrial partners (e.g., feedstock polyol, olefin, aldehyde oxidations). A core component of our technique is the use of advanced microfabricated electrode strategies designed both to enable unsupervised high-throughput materials investigation and advanced generation/collection modes for comprehensive mechanistic understanding. Our existing robotic electrochemistry and physics-informed artificial intelligence methods applied to electrodeposition will result in a robust discovery platform for renewable chemicals. |
| **Rationale for Joint R&D Project** | The focus of the collaboration between the Rodríguez-López and Jackson groups is on developing analytical and artificial intelligence strategies to build accelerated yet robust and rigorous methods that elevate the quality and throughput of electrocatalysis applied to energy and chemical production technologies. We are interested in finding solutions to materials characterization problems we find attractive, but our orientation to fundamental phenomena requires collaboration with industry to take our approach to the next level. We are interested in an R&D partnership for the following reasons: 1) we want to use our fundamental tools to tackle an applied problem with well-defined constraints that is immediately impactful, 2) we are interested in continuing developing our automated electroanalysis platforms tailoring them to industrial standards and statistical methods, 3) we are interested in collaborating with industries that are interested in co-developing AI algorithms that are mutually beneficial to all partners for higher impact. |
| **Keywords** | 1. Electrochemical automation<br>2. Artificial intelligence<br>3. Electrocatalysis<br>4. Hydrogen<br>5. Microelectrodes |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level (2 on our technology, 1 in our hypothesis) | End of the Project | Level (4 after seed and main funding) |
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**R&D Objectives**

***Advanced automated electrochemistry***

The **Rodríguez-López** lab developed and hosts *The Electrolab*[1] and the *eLAB*[2] platforms at the Beckman Institute for Advanced Science and Technology, as part of the Artificial Intelligence in Materials working group. These are advanced electrochemical characterization tools capable of low-supervision experiments at high throughput (HT) with integration of machine learning (ML) and advanced electroanalytical tools (e.g., generation-collection modes, redox feedback, microbands and microelectrodes).[3, 4, 5] In its current form, *The Electrolab*, **Figure 1**, consists of four modules as individual subsystems:1-Python-based software integrating measurement, data analysis, plotting, and simulation libraries into the HardPotato API;[5] 2-multiplexed electrochemical measurement using custom-made microelectrode chips fabricated at UIUC; 3-a moving solvent dispensing and fluid handling robot; and, 4- a printed circuit board (PCB) control platform to integrate all these functions. All modules of *The Electrolab* can be used independent of each other and can be easily incorporated with existing electrochemical instrumentation. The fluidic module consists of an XYZ gantry robot which moves a custom assembly of three 3D-printed nozzles and fluidic-related components (peristaltic pumps, pneumatic pumps, and solenoid valves). The PCB uses an Arduino Mega microcontroller to communicate with the computer via serial to control all hardware. Finally, the multiplexed electrochemistry module is built around microfabricated individually addressable electrode arrays which are fabricated by the **Rodríguez-López** group. These arrays are placed into electrochemical cells and can be individually addressed using a multiplexer. The multiplexer is compatible with any potentiostat (we use a CHI 760E) and is controlled by an onboard Arduino Uno through serial. The use of microfabricated electrode arrays allows us to have multiple pristine electrodes in one electrochemical cell to perform numerous variations in electrochemical conditioning, electrodeposition, electrode characterization, and electrode operation for explorations in electrocatalysis in one single experimental session.

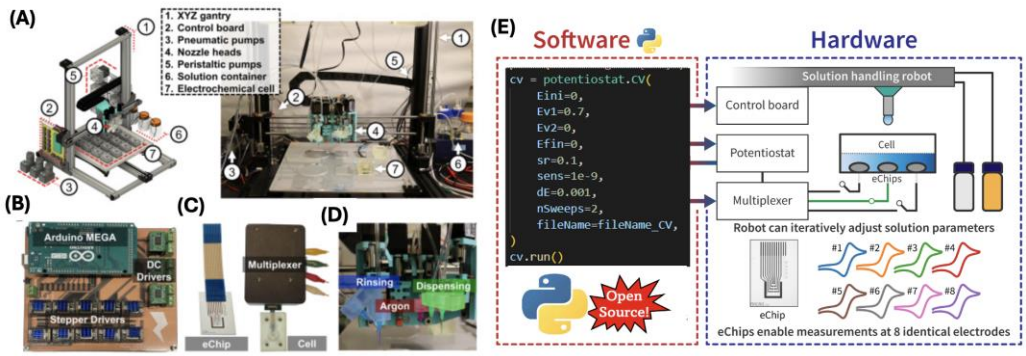


**Figure 1.** The Electrolab and components, including (A) solvent dispensing gantry, (B) integrated electronics, (C) multiplexers and electrochemical chips, (D) a multi-function solvent-dispensing head, and (E) the HardPotato API developed in our group for control.

Beyond *The Electrolab*, the Rodríguez-López lab has developed a simplified solvent handling platform with additional peripheral instrumentation to control pH and the dispensing and gravimetric evaluation of liquid reactants, which we call *eLAB*.[2] The *eLAB*

setup is shown in **Figure 2(A)** where solvent dispensing pumps and valves, stock solutions, a potentiostat, instrumentation for solution measurement and conditioning, and an electrochemical cell capable of using both micro- and macroelectrodes are shown, and schematically organized in **Figure 2(B)**. This setup is versatile and multiple modifications can be done to it for distinct electrocatalytic campaigns. Its simplicity, open source nature (Python API available in our Github) and low cost (~2k USD) allows us to build multiple setups simultaneously, both at Illinois and at our collaboration partners laboratories. The small footprint of *eLAB* has allowed us to install it in a VTI four-port Ar dry box for operation in the absence of oxygen and careful control of solution water content for use in energy storage research, in addition to electrocatalysis. **Figure 2(C)** shows a representative result of the control attainable by *eLAB*, where the kinetics of a redox-mediated electrocatalytic oxidation of isopropanol (IPA) using the electrochemically generated oxidized form of TEMPO in the so-called EC' homogeneous electrocatalysis mechanism was investigated. Precise control of pH, catalyst, and substrate concentration allowed systematic experiments to determine the orders of reaction and the evaluation of the pH dependence on electrocatalytic activity with finesse.

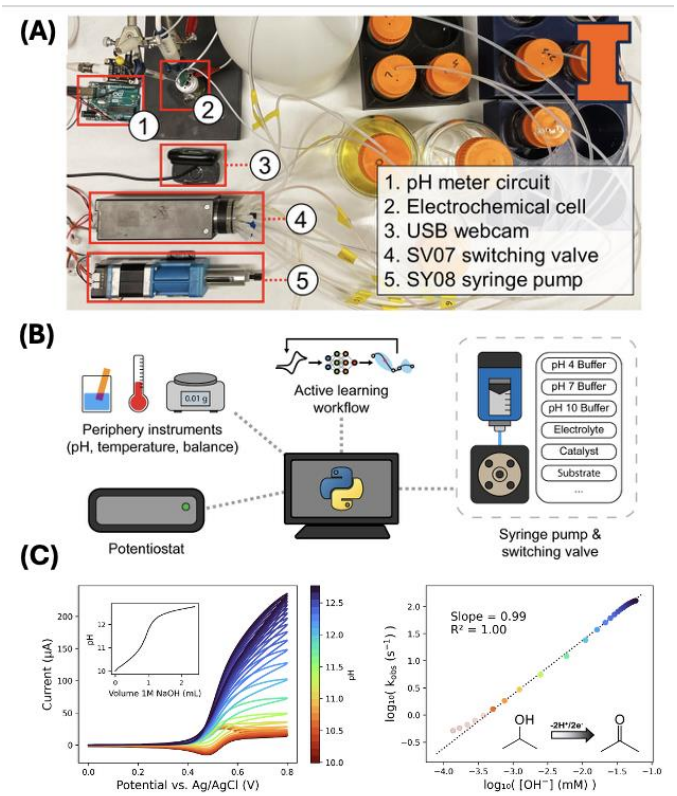


**Figure 2**. Sample configuration of *eLAB* for a campaign involving electrochemical analysis of various TEMPO/alcohol/electrolyte mixtures and involving programmed routines on an electrode. (A) components, (B) block diagram, and (C) example of the pH dependent, automated investigation of isopropanol oxidation to acetone using TEMPO.

Both *eLAB* and the Electrolab can be conveniently controlled to carry out ML-led autonomous explorations, for example using algorithms such as Bayesian Optimization (BO). This is shown in **Figure 3(A)** where *eLAB* was programmed to carry out a closed-loop optimization of TEMPO-catalyzed alcohol oxidations by optimizing for the shape of the voltammogram, with full knowledge that in the EC' mechanism, an optimal and quantifiable response (needed to determine the rate constant for alcohol oxidation) required a sigmoidal-shaped cyclic voltammogram.[3] More specifically, the optimization was done on a multi-

dimensional space, which is typically tackled using a kinetic zone diagram. Regardless of the formalism, in this case we developed the tools to study broad classes of reactions with an optimizable parameter. Critical to demonstrate this capacity, was also the use of a digital twin, using finite element method simulations through COMSOL Multiphysics, as shown in **Figure 3(B)**. In this case, before launching the experimental optimization, we first trained and operated our system in-silico, navigating the kinetic zone diagram so we could understand how noise and non-idealities – captured in 25,000 cyclic voltammograms we simulated – could affect our measurements. We then proceeded to our formal experimental approach, where we tested how quickly we could navigate the kinetic zone diagram to obtain a sigmoid-shape voltammogram, indicative of a kinetically controlled electrocatalytic process, as shown in **Figure 3(C)**. In these cases, the BO process swiftly led to the evolution of the CV from one in which kinetic information cannot be grasped, to one that contained the necessary information to understand the electrocatalytic process, in just a few iterations which unambiguously were superior to a random search. During these iterations, our system autonomously selected the solution system (catalyst and substrate concentrations, scan rate), dispensed it, ran the electrochemistry, cleaned the cell and started again. This demonstrates our ability to couple our system to advanced algorithms in closed-loop.

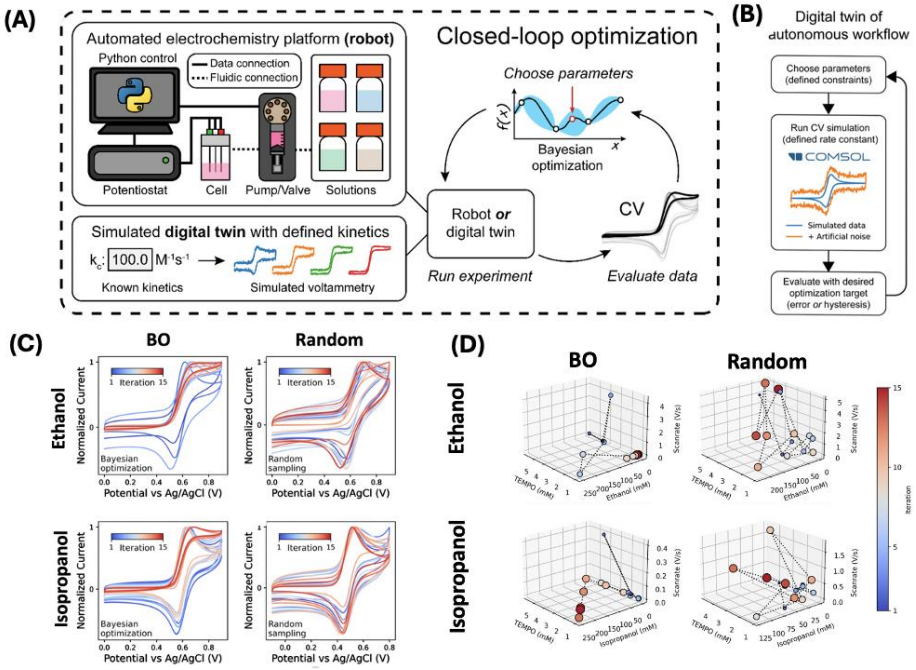


**Figure 3.** Configuration of eLAB for the BO closed-loop experimentation to optimize an electrocatalytic reaction, including (A) procedure combining automated electrochemistry, (B) a digital twin using COMSOL Multiphysics simulations, and (C) and (D) representative results showing the iterations and trajectories taken by the system for arriving to the desired response.

Discovering new heterogeneous catalysts using our platform together with artificial intelligence will require the capacity to electrodeposit metals and evaluate them for their activity. **Figure 4(A)** shows our recent work which demonstrates these capabilities using the electrooxidation of glycerol as a testbed.[6] In this study, we built electrodes that were fully characterized, electrodeposited distinct catalyst surfaces, evaluated their surface activation, tested towards the oxidation of glycerol, and characterized the deposits using SEM. In this case, we designed individually addressable electrodes (IAEs) which consist of microfabricated ships on which we could independently access 8 electrodes; by using several

of these chips, we could not only test new deposits but reproduce them at least three-fold on different chips, to ensure their statistical significance. On each of these electrodes we could then test a new electrocatalyst, and in our case we decided that each catalyst would simply consist of the same electrodeposited metal (Au) but under different deposition potential and time conditions. This work servs as a powerful reminder of the versatility of electrodeposition: our system, allowed us to test how deposition parameters dramatically affect morphology (**Figure 4B, below**) and reactivity (**Figure 4C**), but also it allowed us to identify robust catalysts, i.e., those that had little statistical variability. Indeed, in our work we were able to select the leading deposits – those which showed highest activity at the lowest potential, which we then scaled up to a ~$cm^2$ flow device by electrodepositing over Ni current collectors. Two catalysts were selected, but only one displayed robust operation under flow. This catalyst outperformed our sputtered Au control in all measures and yielded high concentrations of glycolic and lactic acids from the oxidation of glycerol. All our catalyst discovery was driven using our electrochemistry API Hardpotato, controlling each aspect of the electrodeposition and characterization sequence, and performing all the analysis using automated Python routines. This works demonstrates our capabilities for the exploration and discovery of heterogeneous electrocatalysts and its integration to the automated environment developed in our laboratory.[6]

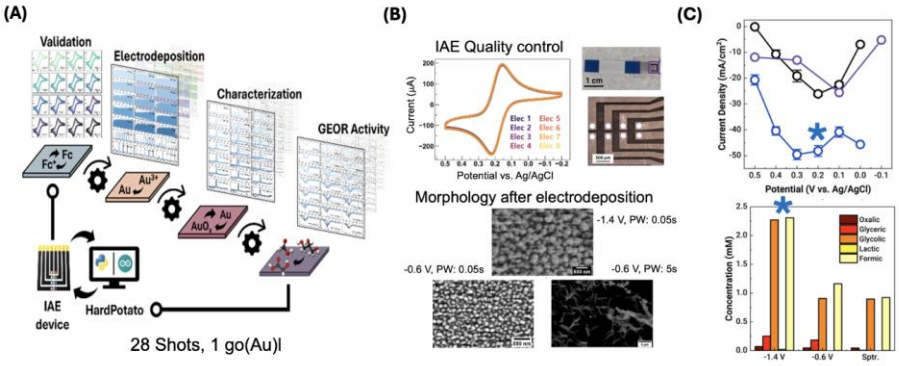


**Figure 4**. Use of HardPotato and individually addressable arrays (IAEs) to electrodeposit, analyze, and characterize Au electrodeposits for the glycerol oxidation reaction; (A) General workflow, (B) Example of IAEs and their characterization, as well as the distinct morphologies of the electrodeposits made under different conditions, and (C) validation in a flow cell of the champion catalysts, showing the down-selection of one catalyst with superior performance.

Finally, we have built sophisticated generation-collection techniques based on electrode designs which are incorporated into the electrochemical automation, **Figure 5**. In these technologies, species that are generated on one electrode, e.g., a catalyst, can be collected onto a neighboring electrode, thus providing key information about the Faradaic efficiency or the mechanism, within the automated environment. We have developed the methods to deploy these complex systems in studies of energy storage, including the measurement of lifetime of molecules for flow batteries and the evaluation of materials for redox-targeted flow batteries.[4] These methods are dramatically simpler than the conventional approach using rotating (ring) disk electrodes or similar, as they do not require moving parts, can be done within the microfabricated chips, and the **Rodriguez-Lopez** group has two-decade long expertise in their design and quantitative characterization. In fact, as shown in **Figure 5(A)**, our group is currently working on the design of low-cost electrodes based on laser-induced graphene technologies, which is allowing us to produce IAEs that integrate generation-collection functions for just a few dollars per chip, and importantly, using a carbon electrode

as the base material for electrodeposition, which is key for the evaluation of electrocatalysts for hydrogen production, and other reactions of interest. Thus, our group is fully equipped and ready to explore new materials for electrocatalysis.

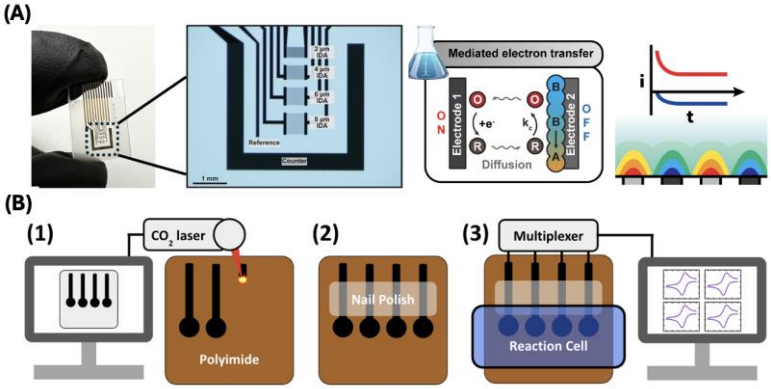


**Figure 5.** Examples of electrode fabrication methods in the Rodríguez-López laboratory, showing (A) an example of interditigated array electrodes for generation-collection methods, depicting a typical electrode structure, its use for interrogating surface intermediates, and its use for determining transient collection of reaction products. (B) Current method for producing low-cost laser-induced graphene electrodes for studies in electrocatalysis.

## Physics informed artificial intelligence methods

In addition to BO methods for closed-loop experiments, Prof. Nick Jackson's group will bring in new AI-guided techniques for promptly identifying superior electrocatalysts, particularly the use of Closed-Loop Transfer (CLT) methods.[7] CLT is a powerful tool enabling AI technology for guiding experimental optimization campaigns, as well as extracting novel and fundamental chemical insights. CLT has been applied to a multi-objective optimization of photostability in a sequence-defined space of chemical oligomers relevant to organic photovoltaics. In a search space of nearly 2,200 possible oligomers, CLT facilitated 1) the improvement of experimentally measured photostability by 500% and 2) the extraction of a novel, validated physical hypothesis after synthesis of only 1.5% of the proposed chemical space (~30 molecules). In addition to the rapid improvement in the optimization of the objective function (experimental photostability), CLT provided the insight that high-energy triplet states dictated the experimental photostability of organic oligomers across chemical space – an insight that was subsequently validated experimentally and represents the first robust and chemically generalizable insight creative by AI to-date. In this project, we will adapt the CLT method to electrochemistry, learning from surface and reactive descriptors to predict electrode surfaces that can accomplish targeted transformations.

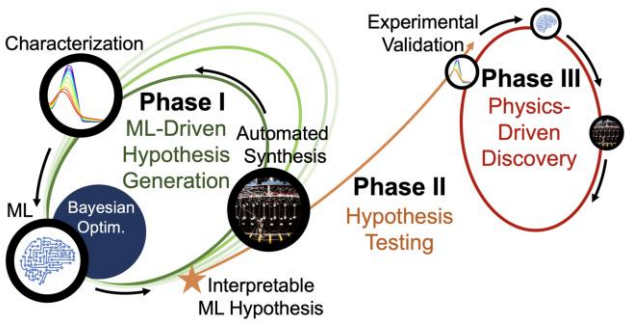


**Figure 6.** Workflow for Closed-Loop Transfer (CLT) methodology showing the ability to propose hypothesis and learn from physics-driven discovery.

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| **R&D Target Specifications** | • The target specification for the electrolysis of any molecule depends on many parameters (concentrations, number of electrons transferred per unit formula, etc.) but in general we are interested in finding electrocatalysts that can deliver transformations at rates that are attractive to their application in fuel cells or electrolyzers, with a benchmark current density of at least ~100 mA/cm$^2$ that can be validated both in electroanalytical and flow cell studies.<br><br>• Another objective of this project is to create a library of materials towards the minimum set of reactive descriptors, from which at least 2-3 catalyst structures of interest are generated for each of the small molecules probed. Ideally, we would be able to predict a working catalyst for a particular molecular by exploring a relatively small number of complementary experiments, typically no more than 2% of the available chemical space, as already described above for CLT. |
| **R&D Plan** | Our project will consist of the following activities organized by stage of funding.<br><br>**6 mo. Seed.**<br>- We will fabricate low-cost laser-induced graphene-based individually addressable arrays (LIG-IAEs) and generation-collection arrays (LIG-GC) using laser printing methods developed in our group (Figure 5). (Months 1, 2)<br>- We will launch a demonstration exploration campaign that will include the construction of a library of catalysts via electrodeposition. We will aim for 4 metals (Au, Pt, Pd) and their combinations in steps of ~30% atomic composition, prepared over at least 30 electrodeposition conditions each, resulting in ~600 unique compositions, each one evaluated in triplicate. (Months 2-5)<br>- We will test autonomously these catalysts towards the HER, HOR, MOR, COR, OER and SOHQ reactions (which includes reactivity to hydrogen, alcohols, CO, and surface oxide processes), to generate a set of measurements to train our system. (Months 2-5)<br>- We will use AI and ML guided designs to test our measurements towards the glycerol oxidation reaction, used here as proxy, for testing our ability to predict the best catalyst towards a small molecule. (Months 2-5)<br>- We will validate our top performers, and a few controls, in lab-scale electrolyzers (Months 4-6)<br>- We will develop the tools to carry out BO-led experiments to optimize certain reactions such as HER for enhancing the production of hydrogen from bespoke electrolytes, particularly using LIG-GC electrodes. (Month 6).<br><br>**Extension to full proposal plan**. Using the seed period as a platform, we would welcome the input from the companies to attack a particular molecule (e.g., an specific organic such as polyol) or a molecule in an environment (e.g., $H_2$ evolution in solutions containing interferences or complex matrices). We would expand our library of catalysts, carry out CLT-based AI methods, and make use of other metals of interest to our collaborators, including their oxides or chalcogenides. |

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## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | All research will be conducted within the Rodríguez-López and Jackson groups, and our groups and members will be the primary responsible for the technical contributions, research scope and securing resources to carry out our tasks. Our groups collaborate within the Artificial Intelligence for Materials (group) that Jackson directs and which is part of the Materials Science and Engineering (MSE) theme at the instate which Rodríguez-López co-leads. The role of the Beckman Institute is to provide space for our collaboration to flourish – this is accomplished by making space for meetings, student office space, and laboratory space which hosts *The Electrolab* and *eLAB* platforms in ~600 $ft^2$ of laboratory space that can be conveniently organized to carry out the activities in our description. In addition, our groups have access to all characterization facilities within the Beckman Institute, Materials Research Laboratory (MRL), and the School of Chemical Sciences (SCS). These already include all sorts of electron microscopy, spectroscopy, X-ray, and other characterization suites, in addition to those in our groups. Our group also has training and access to clean rooms and fabrication facilities around campus. Jackson has access to computational resources as the SCS and within campus. |
| **Expected R&R of Korean SME Partner** | We are primarily interested in our Korean partners to provide 1) Knowledge and guidance on reactive systems that are of their interest, particularly for tackling electrodeposition and eletrocatalytic conversions of interest in non-idealized scenarios, e.g., complex electrolytes, or using particular inputs that help us constraint our experimental explorations and models to those that may be more relevant; 2) guidance on steering our methods towards a particular application – here we have described an approach to tackle a classical problem in electrocatalysis: how to predict a catalyst behavior from an incomplete set of characterization data? – but we are open to other possibilities, as the Rodríguez-López group also works on the field of energy storage (intercalation batteries, redox flow) and other types of mediated electrocatalysis and thermocatalysis; 3) we are interested in learning from our Korean partners which type of ML and AI algorithms they are open to exploring, and which may be compatible with the methods they already used, so we can hone in on such methods for maximum impact. It is also possible that our Korean counterparts have certain materials interests in mind, for which receiving precursors for catalyst synthesis, or molecules for evaluating their electrocatalysis would be appropriate. It is also possible that our Korean counterparts could carry out the validation of our systems in their electrolyzers, which would help inform better our design of materials and help us focus more on the exploration and discovery of materials. |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | • We will design a system and an approach that will help us predict highly active and selective electrocatalysts for broad transformations of interest to the industry from incomplete sets of data and a minimal set of reactions.<br><br>• We will introduce an approach that marries powerful automated and autonomous methods of materials discovery with powerful AI methods to understand the underlying physics and interfacial chemistry.<br><br>• We expect that the methodologies developed could lead to commercialization potential and that the conditions for electrodeposition to generate a distinct catalyst could also lead to intellectual property.<br><br>• The project will help us train new generations of students that could bring this know-how to Korea. Both the Rodríguez-López and Jackson groups have Korean graduate and undergraduate students who could help build a long-term pipeline. |

## 6. References

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